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Abstract

Here goes the abstract written in English.

Resumo

O Resumo fornece ao leitor um sumário do conteúdo da dissertação. Deverá ser breve mas conter detalhe suficiente e, uma vez que é a porta de entrada para a dissertação, deverá dar ao leitor uma boa impressão inicial.

Este texto inicial da dissertação é escrito no fim e resume numa página, sem referências externas, o tema e o contexto do trabalho, a motivação e os objectivos, as metodologias e técnicas empregues, os principais resultados alcançados e as conclusões.

Acknowledgements

The Name of the Author

"You should be glad that bridge fell down.
I was planning to build thirteen more to that same design"

Isambard Kingdom Brunel

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Abbreviations

MIR	Music Information Retrieval
DAW	Digital Audio Workstation
BT	Beat Tracking

Chapter 1

Introdução

O primeiro capítulo da dissertação deve servir para apresentar o enquadramento e a motivação do trabalho e para identificar e definir os problemas que a dissertação aborda. Deve resumir as metodologias utilizadas no trabalho e termina apresentando um breve resumo de cada um dos capítulos posteriores.

1.1 Contexto/Enquadramento

Esta secção descreve a área em que o trabalho se insere, podendo referir um eventual projeto de que faz parte e apresentar uma breve descrição da empresa onde o trabalho decorreu.

1.2 Projeto

Na continuação da secção anterior, e apenas no caso de ser um Projeto e não uma Dissertação, esta secção apresenta resumidamente o projeto.

1.3 Motivação e Objetivos

Apresenta a motivação e enumera os objetivos do trabalho terminando com um resumo das metodologias para a prossecução dos objetivos.

1.4 Estrutura da Dissertação

Introdução

Chapter 2

Literature Review

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Neste capítulo é descrito o estado da arte e são apresentados trabalhos relacionados para mostrar o que existe no mesmo domínio e quais os problemas em aberto. Deve deixar claro que existe uma oportunidade de desenvolvimento que cobre alguma falha concreta.

O capítulo deve também efetuar uma revisão tecnológica às principais ferramentas utilizáveis no âmbito do projeto, justificando futuras escolhas.

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2.1 Introdução

Music Information Retrieval (MIR) is the field of research that focuses on the development of algorithms and systems to analyze, retrieve, and organize music data. MIR bridges a wide range of disciplines, including: musicology, signal processing, machine learning, and human-computer interaction.

============== CONTINUAR

@mullerFundamentalsMusicProcessing2015 ==============

2.2 Visualization Types

We can sum up all music related visualizations into three categories:

1- Sheet Music and Score Images 2- Symbolic Representations 3- Audio Representations

2.2.1 Sheet Music and Score Images

Sheet music describes musical work using a formal language based on musical symbols and letters, thus, this kind of music representations always require a level of music literacy from the performing musician. Furthermore sheet music usally does not induce explicit information on how a given piece should be performed, it is intened for the musician to have a certain level of interpretation liberaty. This can include the variance of tempo, articulation and dynamics amoung others.

Sheet music also comes in a variety of forms that vary throughout History and cultures. For our porpuses we are going to focus on the western standard of music notation. And still, within wester music, there can be vastly differnt forms of notation, however, most are based on a staff which sets five horizontal lines and four spaces, each representing a different musical pitch. Musical relevant symbols are put inside the staff to denote pitch, temporal and expressive indications.

@mullerMusicRepresentations2015a

2.2.2 Symbolic Representations

Symbolic Representations are machine-readable formats where musical events and expressions are explicitly encoded.

These come in various formats, such as MIDI and MusicXML.

Visualy we can identify two main applications for this category, these include: Piano-Roll representations and digital score representations.

Historically piano-roll representations come from so-called **player pianos**, these were self playing pianos that became quite popular in the late 19th to early 20th century. This was achieved by a pneumatic mechanism that would hit keys and pedal of the piano, through a continous roll of paper with perforations. The roll would move over a reading system (known as a tracker bar) that would trigger the musical actions accordingly.

This concept of a continuos and explicit representation of music performance would later prove usefull for digital applications. Such is the case for the Piano-Roll.

The **Piano-Roll** is a stable on any DAW as a visual and interactive tool to write and edit musical events in the form of MIDI notes and atributes (such as velocity and other expressive controls).

MusicXML refers to the XML-based language to transcribe score music into the digital domain. There are multiple sub-varients of the MusicXML format for specific aplications, however all share an XML based langue underneath.

MusicXML was developed to serve as a universal format for storing and sharing music scores across differnt music notation applications. Music notation digital editors then employ proprietary variants of this format for addressing the program's specific charecteristics. In some cases these may not even be xml based, but MusicXML format can be considered the universal format.

These include: .mscz and .mscx - These are Musescore specific formats named for "MuseScore Format" and "Uncompressed MuseScore" format respectively. Both xml based. .dora - "Dorico" specific format (non xml based) .sib - "Sibelius" proprietary format of company Avid (also non xml based) .mei - The "Music Encoding Initiative" developed format for accademic and schollar research. XML based.

2.2.3 Audio Representations

Note: The figures in this chapter will follow the same music exerpt form Choping Op10 n3. They were generated using the AudioVisualizers.py script, which is available in the appendix.

Audio representations are graphs and/or plots that intend to represent sound. These may not be for sctrictly representing music, the intent is to translate sound from the audible to the visual domain. Audio is a term that refers to an acoustic sound that is turned into an electrical signal. Most audio representations, are thus digitaly generated through applying mathematical equations to the audio signal, such as the Fourier Transform.

We can infer at least 3 components that define a given audio. These are, time, frequency and amplitude/energy. An audio visualization will present explicit information of at least 2 of these.

Waveforms a the most comum and one of the simplest ways to represent audio. It consists of a single line on a graph in a 2D graph where amplitude is the y-axis and time is in the x-axis. This representation does not provide frequency information, it only shows how amplitude of the audio changes through it's time.

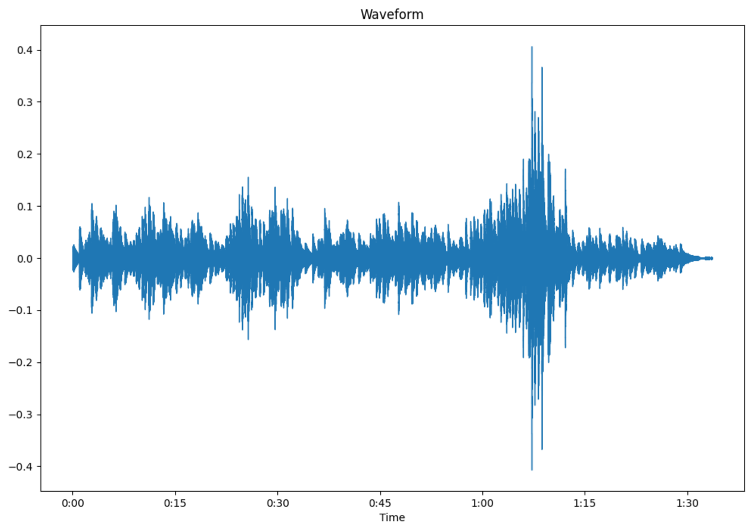

Figure 2.1: Example of an audio waveform.

A spectral visualization refres to a graph with only frequency and amplitude/energy representation in a given time window.

For displaying the three sound components (time, amplitude and frequency/pitch) the graph needs to some how represnt three distict dimensions.

The spectrogram achieves this by having on the y axis the frequency, time on the x-axis and color as substitute for the third axis (amplitude), and in this way displaying all three parameters in a single 2D visual plot. Multiple weights and variances exist of the spectogram that have diferent applications.

The way frequencies are displayed in a given spectogram can help understanding audio better for certain situations. There can be a wide variety of spectogram representations, that can vary in how it's contents are displayed.

2.2.3.1 Spectrogram Variants

The linear spectrogram is the most basic form of spectrogram, where frequencies are displayed in a linear scale. Meaning, frequencies are equally spaced on the y-axis. For music applications, this kind of representation tends to be less useful, as it does not reflect the way humans perceive sound.

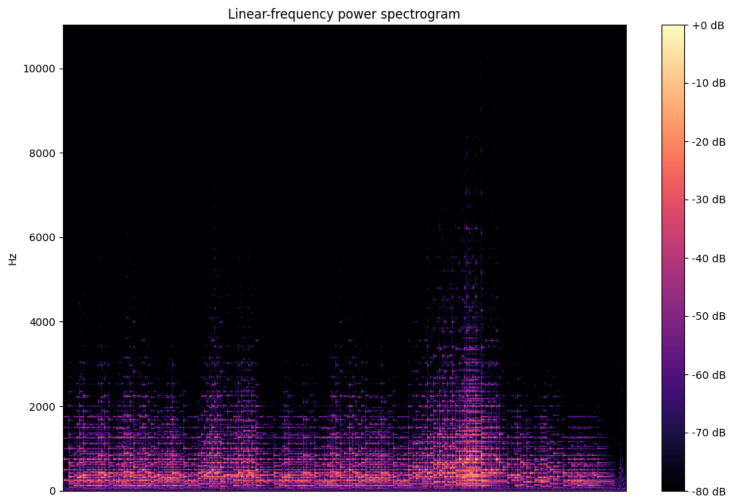

Figure 2.2: Example of Linear spectrogram, content is a musical piece Chopin Op

[Kho, Val, Hen, JKL+, Mü]

2.3 MIR Visualization Tools

2.4 Annotation and Evaluation Tasks in MIR

2.5 Resumo ou Conclusões

Chapter 3

Developed Solution

Este capítulo deve começar por fazer uma apresentação detalhada do problema a resolver1 podendo mesmo, caso se justifique, constituir-se um capítulo separado só com essa finalidade.

Defining the problem: BT, Spectral and Score aligned visualization. Defining the proposed solution: A visualization tool that can present BT and Audio data with an aligned musical score.

Developed solution: Tool for generating custom visualizations that present beat data, spectrogram and score all aligned.

System Architecture: Python based tool, that employs a modular and Object Oriented architecture for generating the visualizations.

Choices made and why

Developed Solution

Chapter 4

Implementation

Este capítulo pode ser dedicado à apresentação de detalhes de nível mais baixo relacionados com o enquadramento e implementação das soluções preconizadas no capítulo anterior. Note-se no entanto que detalhes desnecessários à compreensão do trabalho devem ser remetidos para anexos.

Dependendo do volume, a avaliação do trabalho pode ser incluída neste capítulo ou pode constituir um capítulo separado.

4.1 Developed Iterations

Aqui abordo as diferentes iterações por que passei e a relevancia que cada uma teve para o projeto.

4.2 How to Further Customize Visualizations

4.3 Summary

Chapter 5

Conclusões e Trabalho Futuro

Deve ser apresentado um resumo do trabalho realizado e apreciada a satisfação dos objetivos do trabalho, uma lista de contribuições principais do trabalho e as direções para trabalho futuro.

A escrita deste capítulo deve ser orientada para a total compreensão do trabalho, tendo em atenção que, depois de ler o Resumo e a Introdução, a maioria dos leitores passará à leitura deste capítulo de conclusões e recomendações para trabalho futuro.

5.1 Achievements

Concretizações: Provar a necessidade de explorar visualizações que incorporam partituras alinhadas com tempo real de uma performance/gravação. Provar o valor de usar uma ferramenta modular em python.

5.2 Future Work

Problemas da solução: O problema do sistema estar dependente do SVG no formato em que sai do ScoreWarp. Dificuldade em obter os ficheiros correctos (MAPS, MEI, etc) O problema de nao conseguir definir o espaçamento da partitura. O problema da representação do espetrograma como PNG. Trabalho Futuro: Integração com um GUI. Problema da paginação Integração para visualização de outros parametros MIR. Integração de metodos para gerar visualizações costumizaveis sem ter de alterar manualmente o SVG.

References

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REFERENCES

Appendix A

Loren Ipsum

Depois das conclusões e antes das referências bibliográficas, apresenta-se neste anexo numerado o texto usado para preencher a dissertação.

A.1 O que é o *Loren Ipsum*?

Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining essentially unchanged. It was popularised in the 1960s with the release of Letraset sheets containing Lorem Ipsum passages, and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum [**?**].

A.2 De onde Vem o Loren?

Contrary to popular belief, Lorem Ipsum is not simply random text. It has roots in a piece of classical Latin literature from 45 BC, making it over 2000 years old. Richard McClintock, a Latin professor at Hampden-Sydney College in Virginia, looked up one of the more obscure Latin words, consectetur, from a Lorem Ipsum passage, and going through the cites of the word in classical literature, discovered the undoubtable source. Lorem Ipsum comes from sections 1.10.32 and 1.10.33 of "de Finibus Bonorum et Malorum" (The Extremes of Good and Evil) by Cicero, written in 45 BC. This book is a treatise on the theory of ethics, very popular during the Renaissance. The first line of Lorem Ipsum, "Lorem ipsum dolor sit amet...", comes from a line in section 1.10.32.

The standard chunk of Lorem Ipsum used since the 1500s is reproduced below for those interested. Sections 1.10.32 and 1.10.33 from "de Finibus Bonorum et Malorum" by Cicero are also reproduced in their exact original form, accompanied by English versions from the 1914 translation by H. Rackham.

A.3 Porque se usa o Loren?

It is a long established fact that a reader will be distracted by the readable content of a page when looking at its layout. The point of using Lorem Ipsum is that it has a more-or-less normal distribution of letters, as opposed to using “Content here, content here”, making it look like readable English. Many desktop publishing packages and web page editors now use Lorem Ipsum as their default model text, and a search for “lorem ipsum” will uncover many web sites still in their infancy. Various versions have evolved over the years, sometimes by accident, sometimes on purpose (injected humour and the like).

A.4 Onde se Podem Encontrar Exemplos?

There are many variations of passages of Lorem Ipsum available, but the majority have suffered alteration in some form, by injected humour, or randomised words which don’t look even slightly believable. If you are going to use a passage of Lorem Ipsum, you need to be sure there isn’t anything embarrassing hidden in the middle of text. All the Lorem Ipsum generators on the Internet tend to repeat predefined chunks as necessary, making this the first true generator on the Internet. It uses a dictionary of over 200 Latin words, combined with a handful of model sentence structures, to generate Lorem Ipsum which looks reasonable. The generated Lorem Ipsum is therefore always free from repetition, injected humour, or non-characteristic words etc.